



Some Remarks on Replicated Simulated Annealing

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Received: 7 October 2020 / Accepted: 13 February 2021 / Published online: 2 March 2021
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Abstract

Recently authors have introduced the idea of training discrete weights neural networks using a mix between classical simulated annealing and a replica ansatz known from the statistical physics literature. Among other points, they claim their method is able to find robust configurations. In this paper, we analyze this so called “replicated simulated annealing” algorithm. In particular, we give criteria to guarantee its convergence, and study when it successfully samples from configurations. We also perform experiments using synthetic and real data bases.

Keywords Simulated annealing · Replicas · Perceptron · Neural networks · Optimization

Mathematics Subject Classification 60J10 · 82C32 · 65C05

1 Introduction

In the past few years, there has been a growing interest in finding methods to train discrete weights neural networks. As a matter of fact, when it comes to implementations,

Communicated by Hal Tasaki.

The research of the second author was funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany’s Excellence Strategy EXC 2044-390685587, Mathematics Münster: Dynamics—Geometry—Structure.

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